

Exercises: Simple Gaussian Linear Regression Part III

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(Referring to the theoretical parts: 4, 5, 6, 7, 8)

1 Computer repair data

A corporation sells computer parts and performs maintenance and repair service. The data below have been collected from 18 recent calls to perform maintenance service; for each call, x is the number of repairs and y is the total number of minutes spent by the service person. Assume that a Gaussian linear model $Y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$ is appropriate. Knowing that:

$$\begin{aligned} \sum_{i=1}^{18} y_i &= 1152 & \sum_{i=1}^{18} x_i &= 81 \\ \sum_{i=1}^{18} (y_i - \bar{y})^2 &= 16504 & \sum_{i=1}^{18} (x_i - \bar{x})^2 &= 74.5 & \sum_{i=1}^{18} (y_i - \bar{y})(x_i - \bar{x}) &= 1098 \end{aligned}$$

where $\bar{x}$ and $\bar{y}$ are the sample means, and

$$\sum_{i=1}^{18} (y_i - \hat{y}_i)^2 = 321.39$$

- Obtain the estimated regression function.
- Interpret β_2 in your estimated regression function. Does β_1 provide any relevant information here? Explain.
- Perform a statistical test with a system of hypothesis $H_0 : \beta_2 = 0$ vs. $H_1 : \beta_2 > 0$. What can we conclude at 5% significance level?
- Compute a 95% confidence interval for β_2 .
- Test the hypothesis $H_0 : \beta_2 = 0$ vs. $H_1 : \beta_2 \neq 0$ at a significance level 0.05. Was it necessary to perform this test, given what you already knew from the previous points? Explain.

- f) Obtain a point estimate of the mean service time when $x = 5$ parts are repaired.
Obtain a confidence interval with level 0.90.
- g) Obtain the partition of the total sum of squares and coefficient R^2 .

1 Computer repair data

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$$\sum_{i=1}^{18} y_i = 1152 \qquad \sum_{i=1}^{18} x_i = 81$$
$$\sum_{i=1}^{18} (y_i - \bar{y})^2 = 16504 \qquad \sum_{i=1}^{18} (x_i - \bar{x})^2 = 74.5 \qquad \sum_{i=1}^{18} (y_i - \bar{y})(x_i - \bar{x}) = 1098$$

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- e) Test the hypothesis $H_0 : \beta_2 = 0$ vs. $H_1 : \beta_2 \neq 0$ at a significance level 0.05. Was it necessary to perform this test, given what you already knew from the previous points? Explain.

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- f) Obtain a point estimate of the mean service time when $x = 5$ parts are repaired. Obtain a confidence interval with level 0.90.
- g) Obtain the partition of the total sum of squares and coefficient R^2 .

$$n = 18$$

$$\sum y_i = 1152$$

$$\sum x_i = 81$$

$$SST = \sum (y_i - \bar{y})^2 = 16504$$

$$\sum (y_i - \bar{y})(x_i - \bar{x}) = 1098$$

$$\sum (x_i - \bar{x})^2 = 74.5$$

$$SSE = \sum (y_i - \hat{y}_i)^2 = 321.39$$

ASSUMPTIONS

- $E(\varepsilon_i) = 0 \quad \forall i \rightarrow E(y_i) = \hat{\beta}_1 + \hat{\beta}_2 x_i$
 - $Cov(\varepsilon_i, \varepsilon_k) = 0 \quad \text{for } i \neq k \rightarrow Cov(y_i; y_k) = 0 \quad \text{for } i \neq k$
 - $VAR(\varepsilon_i) = \sigma^2 > 0$
 - FIXED AND NOT STOCHASTIC
 - NOT ALL EQUAL $x_1, \dots, x_{18}$
 - $\varepsilon_i \sim N(0, \sigma^2)$

$$y_i = \hat{\beta}_1 + \hat{\beta}_2 x_i$$

$$\bar{y} = \frac{1}{n} \sum y_i = \frac{1152}{18} = 64$$

$$\bar{x} = \frac{1}{n} \sum x_i = \frac{81}{18} = 4.5$$

- $\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x} = 64 - 4.5 \cdot 14.738 = -2.321$

- $\hat{\beta}_2 = \frac{S_{xy}}{S_x^2} = \frac{\frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n-1} \sum (x_i - \bar{x})^2} = \frac{1098}{74.5} = 14.738$

MINUTE SPEND

$$y_i = -2.321 + 14.738 x_i$$

NOT IMPORTANT, NO CALL IS -2.321 MIN LONG

OF REPAIRS

WHEN KEEPING EVERYTHING CONSTANT, IF WE INCREASE THE NUMBER OF REPAIR BY 1, THE TIME INCREASE OF 14 MIN

$$\begin{cases} H_0: \beta_2 = 0 & \alpha = 0.05 \\ H_1: \beta_2 > 0 \end{cases}$$

$$t_2^{OBS} = \frac{\hat{\beta}_2 - \beta_2}{\sqrt{VAR(\hat{\beta}_2)}} = \frac{14.738 - 0}{\sqrt{0.253}} = 29.25$$

$$VAR(\hat{\beta}_2) = \frac{s^2}{\sum (x_i - \bar{x})^2} = \frac{\frac{1}{n-p} SSE}{74.5} = \frac{\frac{1}{18-2} 321.39}{74.5} = 0.253$$

$$\alpha^{OBS} = P(|t_{n-p}| \geq |t_2^{OBS}|) = \underset{\substack{\uparrow \\ \text{1-TAIL}}}{\min} \left\{ P(t_{n-p} \leq -t_2^{OBS}); P(t_{n-p} > t_2^{OBS}) \right\}$$

$$\alpha^{OBS} < 1 - 0.9995 \approx 0$$

$$\begin{array}{c} \alpha^{OBS} \qquad \qquad \alpha^{OBS} \\ | \quad \quad \quad | \\ R \quad \quad NO P. \end{array}$$

$$\text{SINCE } \alpha^{OBS} < \alpha = 0.05 \rightarrow \text{REJECT } H_0$$

95% CONFIDENCE INTERVAL FOR β_2

$$CI(0.95) = \left[\hat{\beta}_2 - t_{n-2; 1-\alpha/2} \sqrt{VAR(\hat{\beta}_2)}; \hat{\beta}_2 + t_{n-2; 1-\alpha/2} \sqrt{VAR(\hat{\beta}_2)} \right]$$
$$\left[14.738 - 2.12 \cdot 0.50; 14.738 + 2.12 \cdot 0.50 \right]$$
$$\left[13.638; 15.938 \right]$$

$$n-p = 18-2 = 16$$

$$1-\alpha = 0.95 \rightarrow \alpha = 0.05 \rightarrow 1-\alpha/2 = 0.975$$

$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$$

$$\alpha = 0.05$$

$$t_2^{OBS} = \frac{\hat{\beta}_2 - \beta_2}{\sqrt{VAR(\hat{\beta}_2)}} = 29.25$$

$$\alpha^{OBS} = P(|t_{n-p}| \geq |t_2^{OBS}|) = 2 \min \left\{ P(t_{n-p} \leq -t_2^{OBS}); P(t_{n-p} > t_2^{OBS}) \right\} = 2(1 - 0.9995) = 0.01$$

α^{obs}
 $\downarrow$
 R NO R

SINCE $\alpha^{obs} < \alpha = 0.05 \rightarrow \text{REJ } H_0$

$$\hat{y}_i^* = -2.321 + 11.738 x_i^* = 71.369$$

↑
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$$\hat{y}^* \sim N(\underbrace{\beta_1 + \beta_2 x^*}_{\mu^*}; \frac{\sigma^2}{n} + \frac{(x^* - \bar{x})^2}{\sum (x_i - \bar{x})^2} \sigma^2)$$

$$\sqrt{\frac{\frac{\sigma^2}{n} + \frac{(x^* - \bar{x})^2}{\sum (x_i - \bar{x})^2} \sigma^2}{\frac{\sigma^2}{n} + \frac{(x^* - \bar{x})^2}{\sum (x_i - \bar{x})^2} \sigma^2}} = s \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

$$(\hat{y}^* - t_{n-2; 1-\alpha/2} s \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum (x_i - \bar{x})^2}}; \hat{y}^* + t_{n-2; 1-\alpha/2} s \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum (x_i - \bar{x})^2}})$$

$$s^2 = \frac{SSE}{n-p} = \frac{321.39}{18-2} = 20$$

$$s = \sqrt{s^2} = 4.472$$

$$t_{n-2; 1-\alpha/2} = t_{16; 0.95} = 1.746$$

$$\frac{(x^* - \bar{x})^2}{\sum (x_i - \bar{x})^2} = \frac{(5 - 4.5)^2}{79.5} = 0.0031$$

↑ FROM THE DATA

$$CI(0.95) = (69.482; 73.258)$$

$$R^2 = \frac{SSR}{SST} = \frac{SST - SSE}{SST} = 0.98$$

$$SST = 165.04$$

$$SSE = 321.39$$